Vulnerability Assessment and Firewall Configuration Using pfSense

Title:

Implementation and Validation of Firewall Rules Using pfSense in a Virtual lab Environment

1. Introduction

This lab focused on building a secure virtual network and implementing firewall controls using pfSense. The objective was to configure a controlled environment where only specific services are accessible while all other network traffic is restricted. This simulates real-world cybersecurity practices such as network segmentation, access control, and attack surface reduction.

2. Objectives

- Configure a virtual lab environment using multiple machines
- Install and configure pfSense as a firewall
- Deploy a web server and simulate a database service
- Create firewall rules to allow and block specific traffic
- Validate configurations using Nmap scanning

Component	Description
Firewall	pfSense 2.8.1 (VirtualBox VM)
Client/Server	Windows 10 VM
Attacker	Kali Linux VM
Platform	Oracle VirtualBox
Network Type	Internal Network (LAN) + NAT (WAN)

4. Methodology

4.1 pfSense Installation and Configuration

- Downloaded pfSense ISO from Netgate
- Installed pfSense on a VirtualBox VM
- Configured:
  - WAN interface → DHCP (NAT network)
  - LAN interface → Static (192.168.56.1/24)

4.2 Windows Server Setup

- Configured Windows 10 VM within the LAN network
- Verified connectivity with pfSense
- Installed **Internet Information Services (IIS)** to host a web server
- Verified access via http://localhost

4.3 Simulated Database Service

- Since no SQL Server was installed, a listening port was created using PowerShell:

```
$listener = [System.Net.Sockets.TcpListener]1433
$listener.Start()
```

- Verified port status:

```
netstat -an | findstr 1433
```

4.4 Firewall Rule Configuration (pfSense)

Firewall rules were configured under:

Firewall → Rules → LAN

Rule	Action	Protocol	Destination	Port
1	Allow	TCP	192.168.56.107	80 (HTTP)
2	Allow	TCP	192.168.56.107	1433 (SQL)

Rule	Action	Protocol	Destination	Port
3	Block	TCP	192.168.56.107	7680
4	Block	Any	Any	Any

4.5 Validation Using Nmap

From Kali Linux:

nmap -p- 192.168.56.107

5. Results

Nmap Scan Output:

80/tcp open http
1433/tcp open ms-sql-s
Not shown: 65533 filtered ports

6. Analysis

The results confirm that:

- **Port 80 (HTTP)** is open → Web server is accessible
- **Port 1433 (SQL)** is open → Database service is reachable
- **All other ports are filtered** → Firewall is effectively blocking unauthorized access

This demonstrates that the pfSense firewall is correctly enforcing access control policies and limiting exposure only to required services.

7. Key Concepts Demonstrated

- Network segmentation
- Firewall rule creation and prioritization
- Service exposure and port management
- Traffic filtering and access control

- Network validation using Nmap

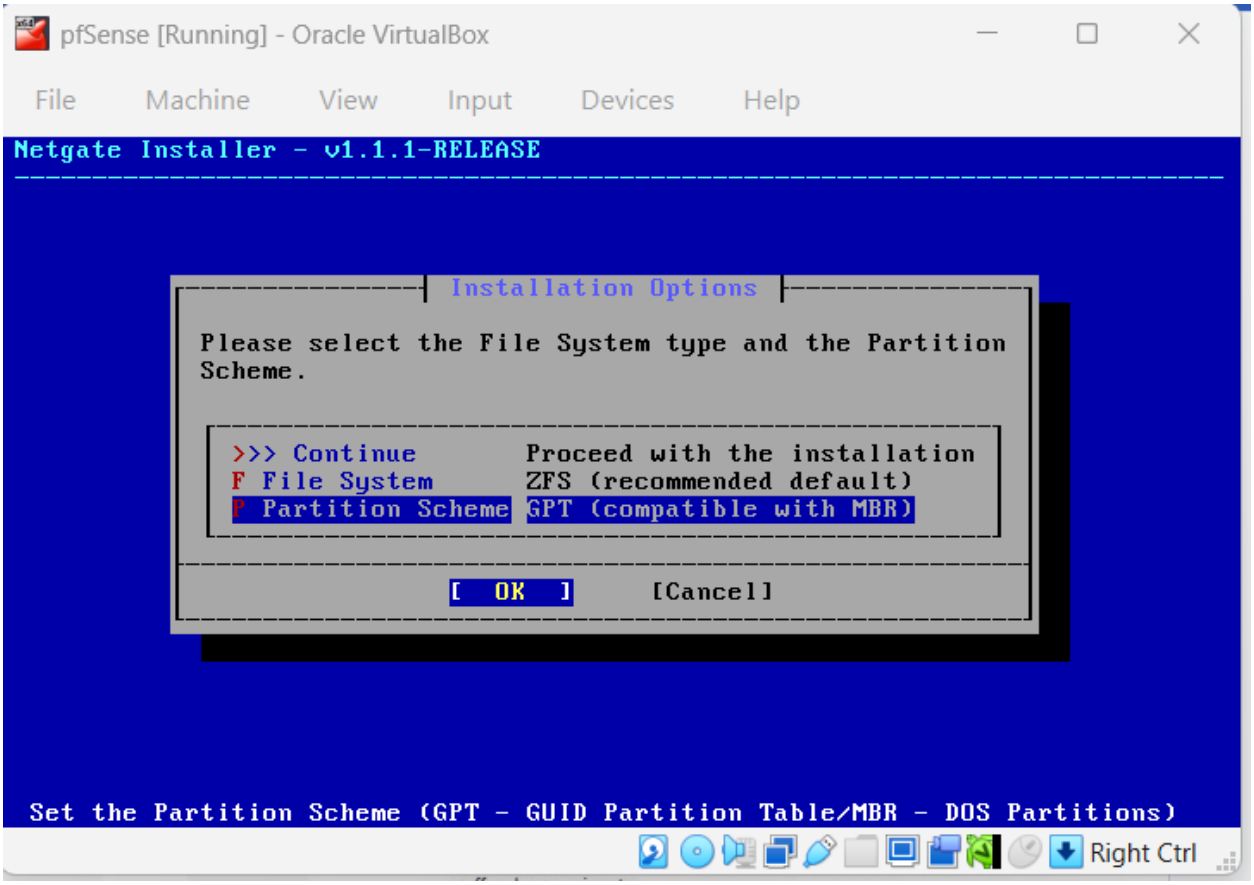
8. Challenges Encountered

- Initial inability to access pfSense GUI due to IP mismatch
- Port 1433 not appearing open because no service was running
- Firewall rule ordering causing unintended blocking
- Windows firewall requiring elevated privileges for rule creation

These issues were resolved through troubleshooting and proper configuration adjustments.

SCREENSHOTS:

Downloading and Installing pfSense:



pfSense [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Installation Details

```
Exporting the network settings to pfSense.

Selected file system type and partition scheme: ZFS and GPT.

ZFS Pool Name: pfSense.

Installing pkg:

pkg-static: Warning: Major OS version upgrade detected.  Running "pkg boot
Updating pfSense-core repository catalogue...
Fetching meta.conf: . done
Fetching data.pkg: . done
Processing entries: . done
pfSense-core repository update completed. 4 packages processed.
Updating pfSense repository catalogue...
Fetching meta.conf: . done
Fetching data.pkg: .......... done
Processing entries:
Processing entries............. done
pfSense repository update completed. 541 packages processed.
All repositories are up to date.
```

Right Ctrl

pfSense [Running] - Oracle VirtualBox

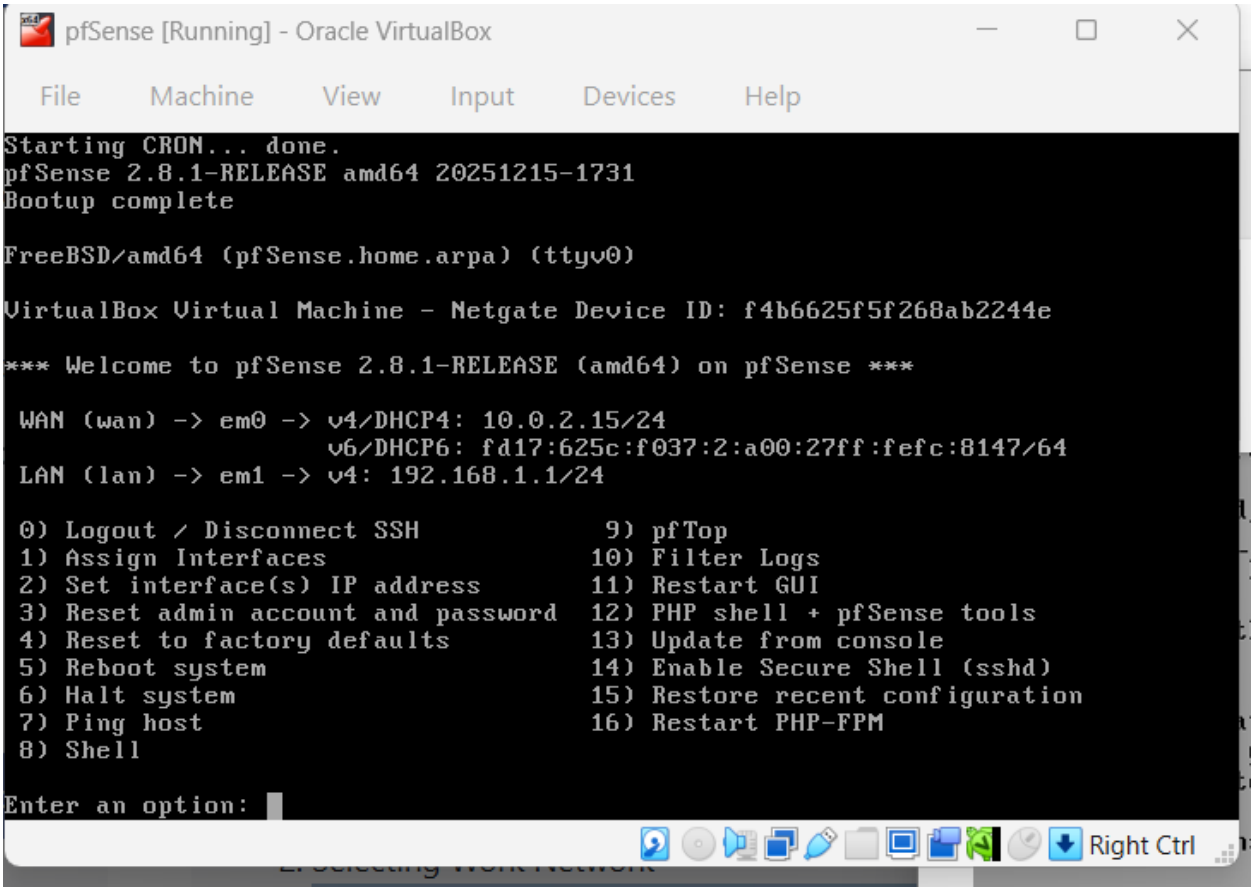
File Machine View Input Devices Help

Installation Details

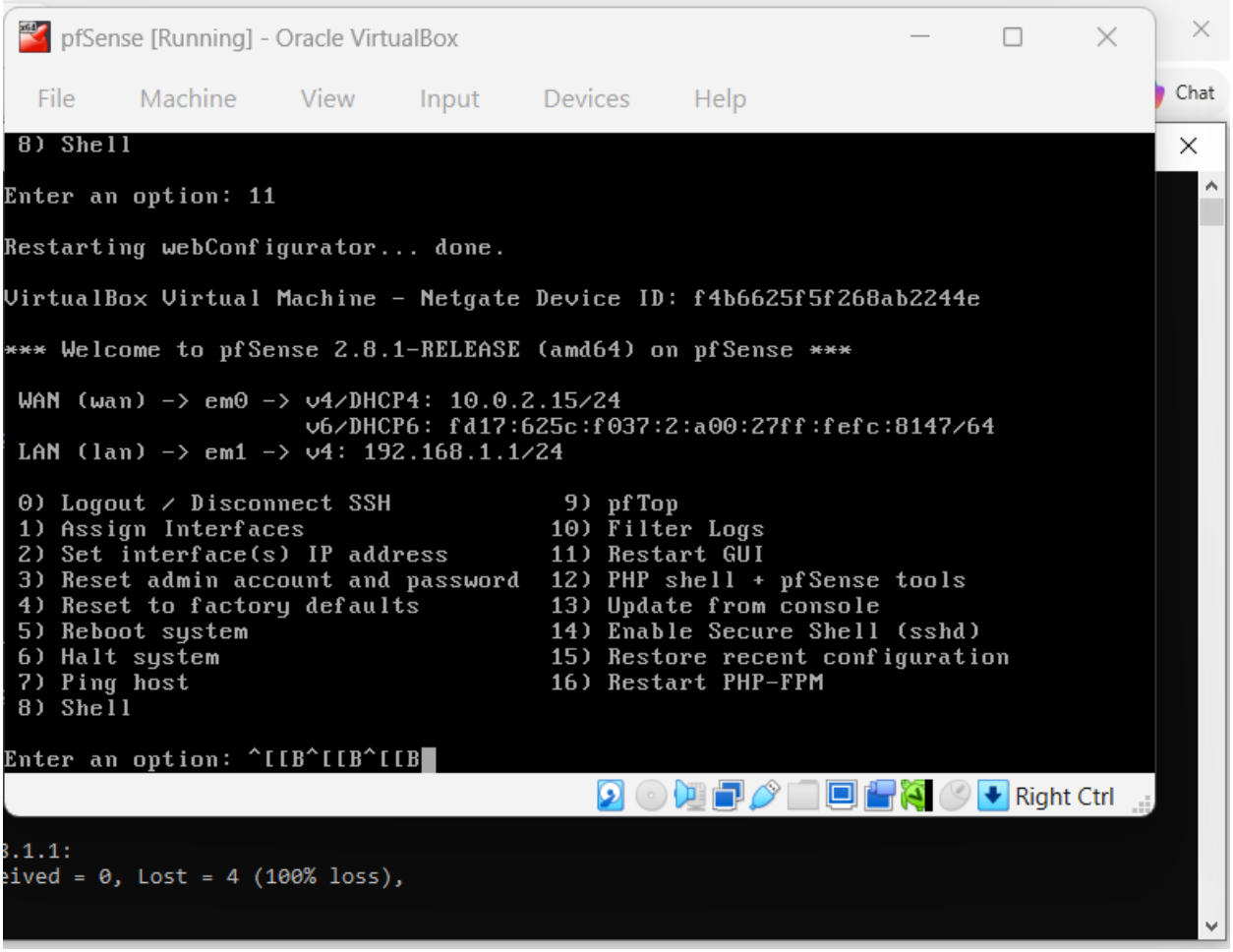
```
/etc/rc.conf. radeonkms for older AMD GPUs can be loaded and there are
some positive reports if EFI boot is NOT enabled.

For amdgpu: kld_list="amdgpu"
For Intel: kld_list="i915kms"
For radeonkms: kld_list="radeonkms"

Please ensure that all users requiring graphics are members of the
"video" group.

Please note that this package was built for FreeBSD 15.0.
If this is not your current running version, please rebuild
it from ports to prevent panics when loading the module.

pfSense Post Installation setup

Exporting the network settings to pfSense.

pfSense Post Installation setup .. done.
```

< OK >

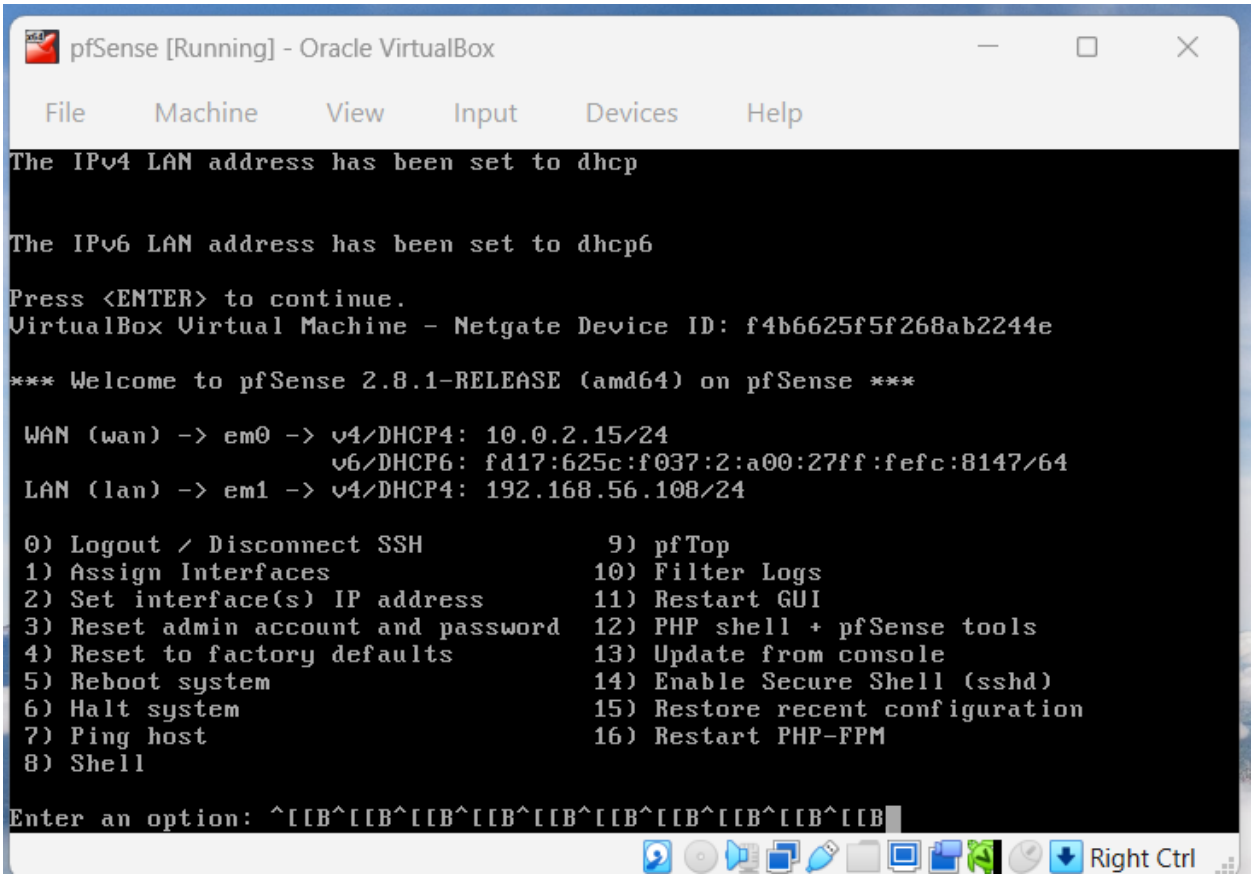
Right Ctrl


pfSense [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20251215-1731
Bootup complete
FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)
VirtualBox Virtual Machine - Netgate Device ID: f4b6625f5f268ab2244e
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***
WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
v6/DHCP6: fd17:625c:f037:2:a00:27ff:fefc:8147/64
LAN (lan) -> em1 -> v4: 192.168.1.1/24
0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM
Enter an option:
Right Ctrl

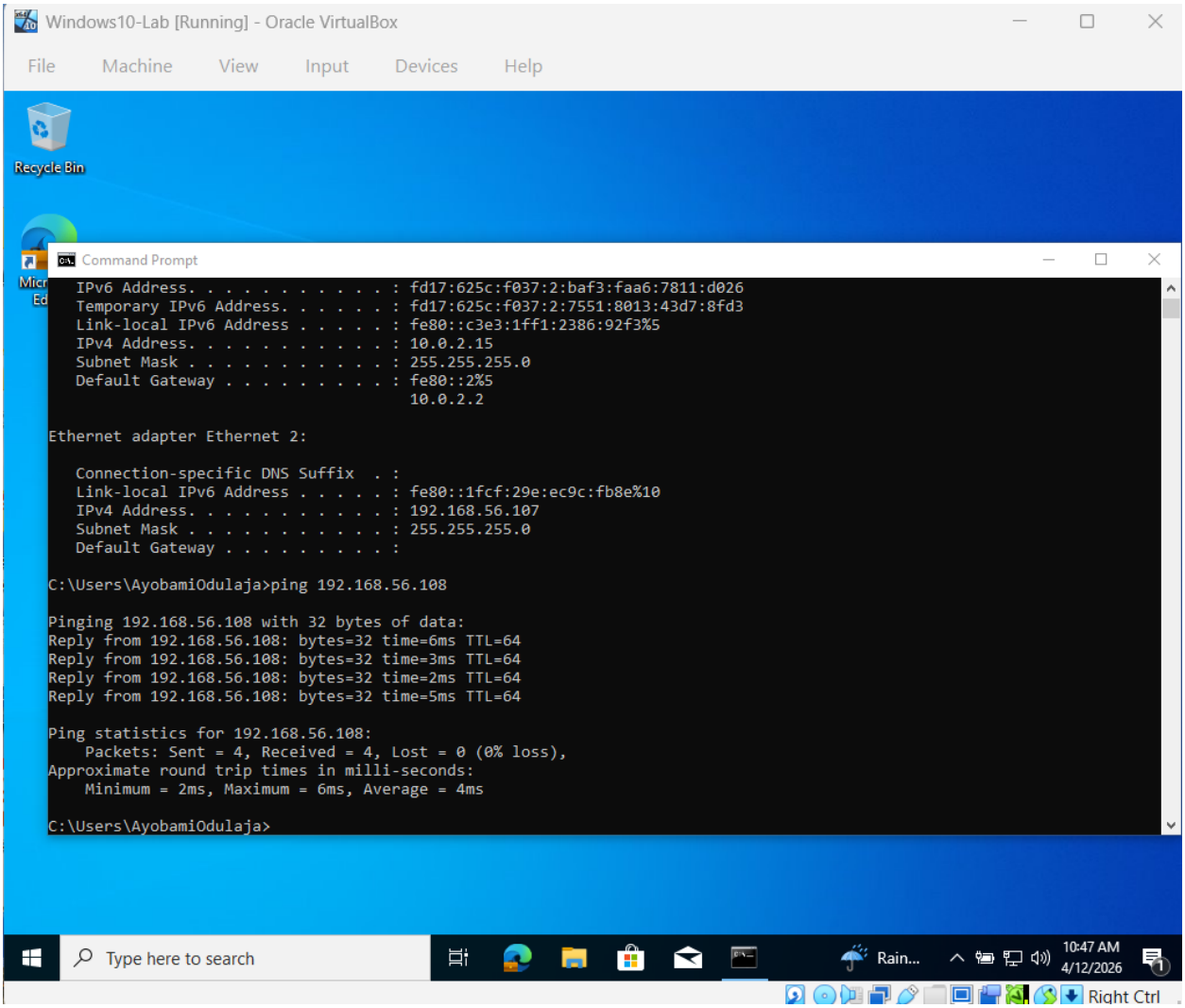
Restarted GUI



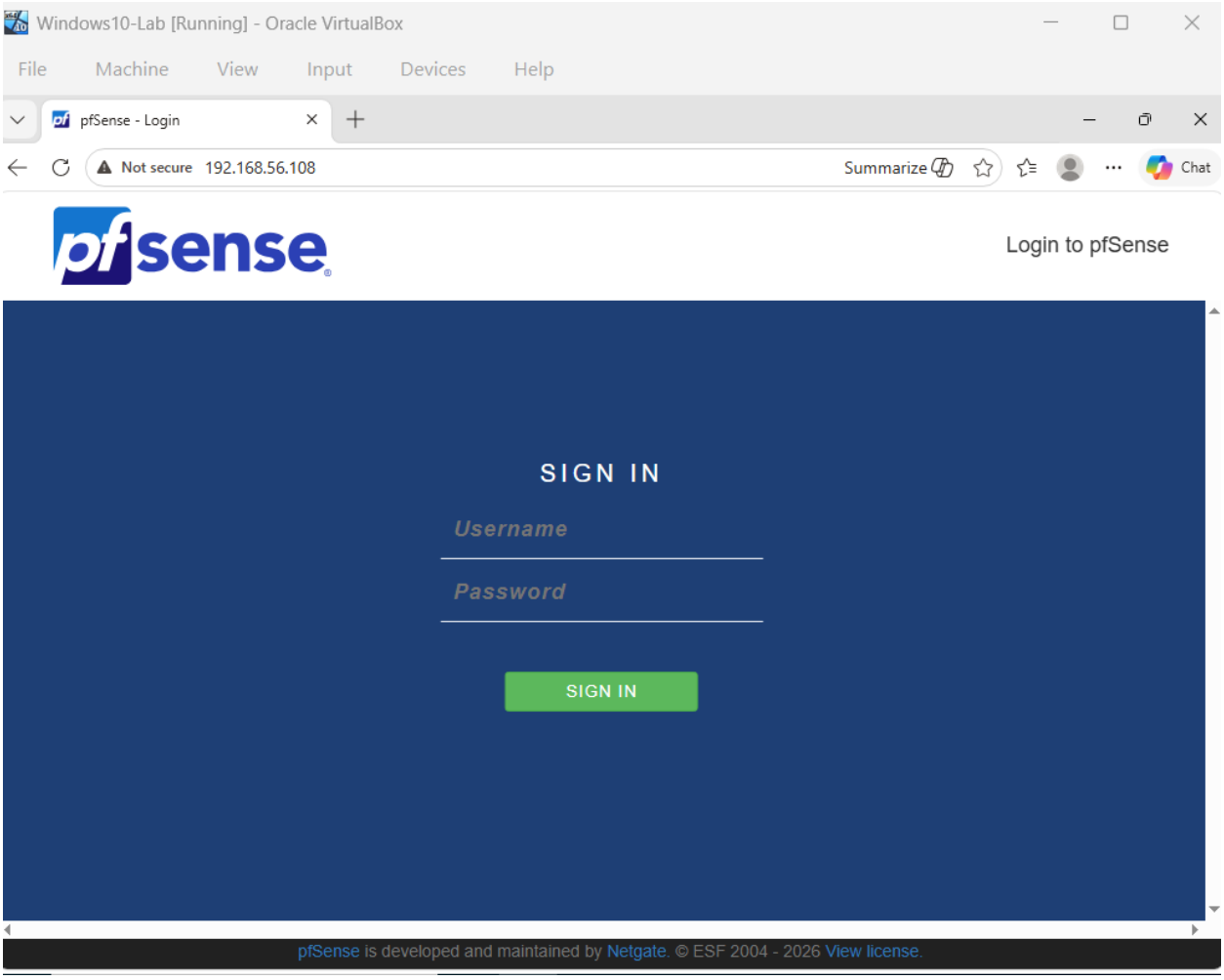
Configuring interface of pfSense:



Ping pfSense via window VM:



pfSense GUI:


Windows10-Lab [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
pfSense - Login
Not secure
192.168.56.108
Summarize
Chat
pfsense
Login to pfSense
SIGN IN
Username
Password
SIGN IN
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Windows10-Lab [Running] - Oracle VirtualBox

File Machine View Input Devices Help

pfSense.home.arpa - Wizard: pfSe

Not secure 192.168.56.108/wizard.php?xml=setup_wizard.xml

pfsense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.

Wizard / pfSense Setup /

pfSense Setup

Welcome to pfSense® software!

This wizard will provide guidance through the initial configuration of pfSense.

The wizard may be stopped at any time by clicking the logo image at the top of the screen.

pfSense® software is developed and maintained by Netgate®

Learn more

Next

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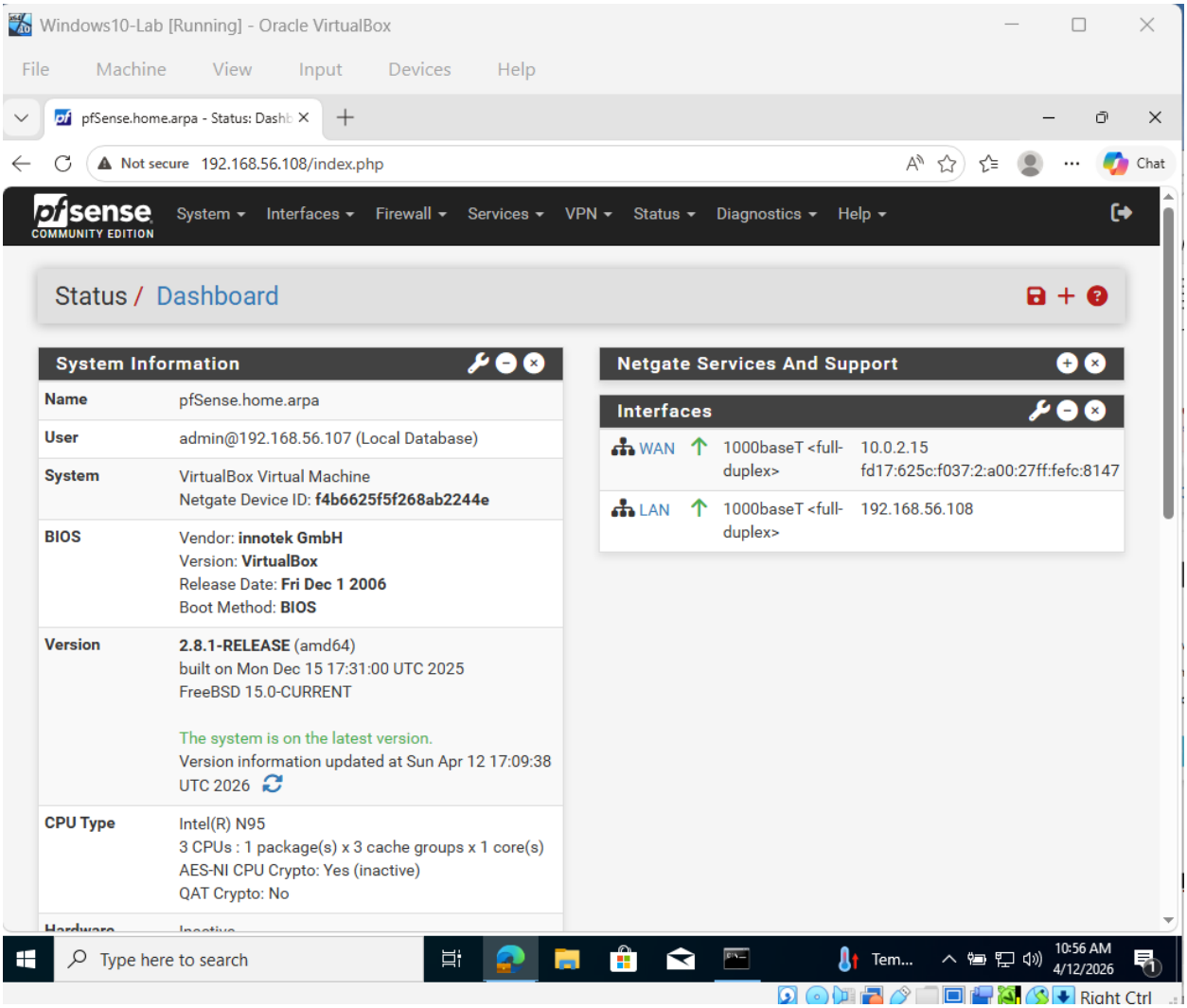
Dashboard:

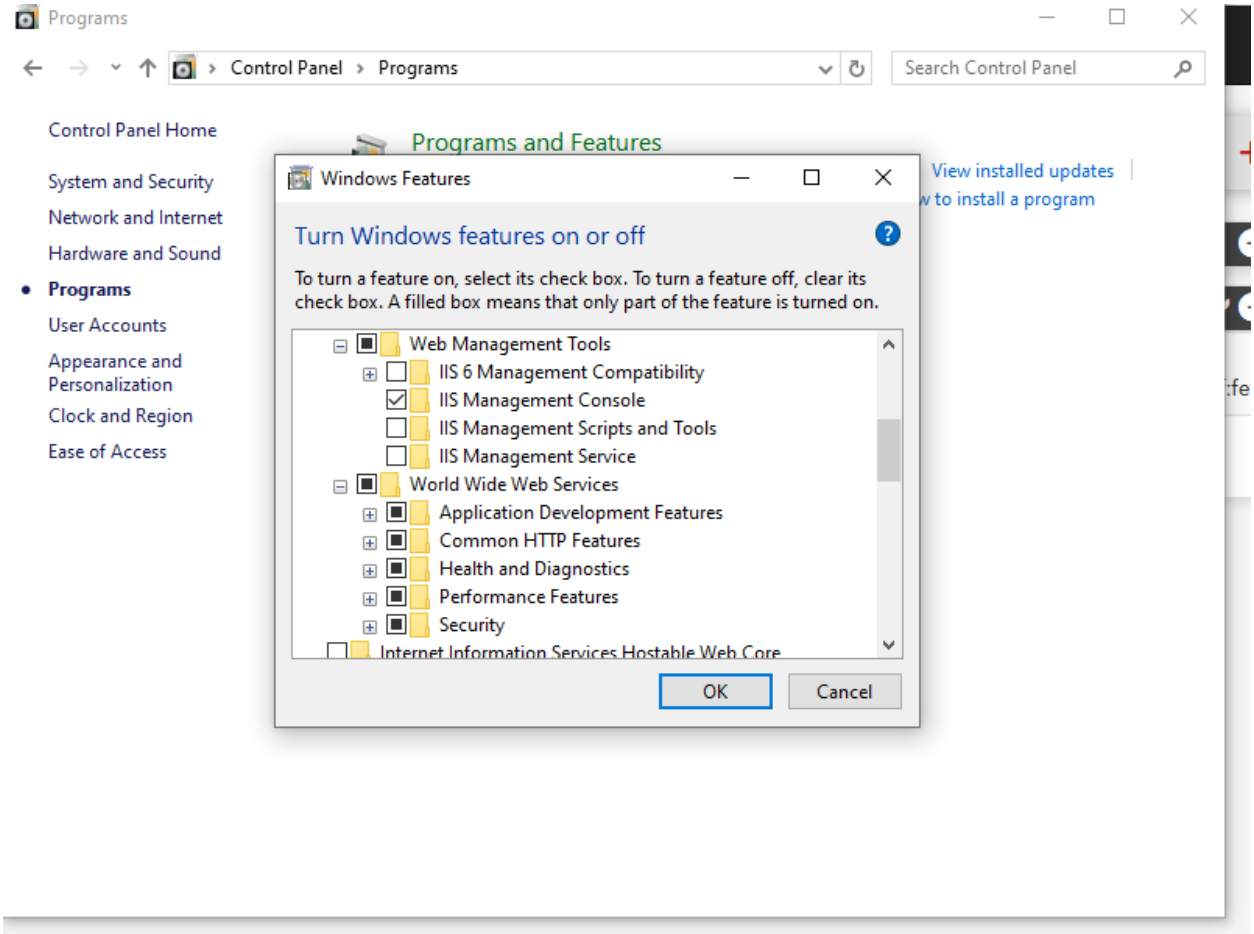

Windows10-Lab [Running] - Oracle VirtualBox
File Machine View Input Devices Help
pfSense.home.arpa - Status: Dashb
Not secure 192.168.56.108/index.php
Chat
pfsense COMMUNITY EDITION
System Interfaces Firewall Services VPN Status Diagnostics Help
Status / Dashboard
System Information
Name pfSense.home.arpa
User admin@192.168.56.107 (Local Database)
System VirtualBox Virtual Machine
Netgate Device ID: f4b6625f5f268ab2244e
BIOS Vendor: innotek GmbH
Version: VirtualBox
Release Date: Fri Dec 1 2006
Boot Method: BIOS
Version 2.8.1-RELEASE (amd64)
built on Mon Dec 15 17:31:00 UTC 2025
FreeBSD 15.0-CURRENT
The system is on the latest version.
Version information updated at Sun Apr 12 17:09:38 UTC 2026
CPU Type Intel(R) N95
3 CPUs : 1 package(s) x 3 cache groups x 1 core(s)
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No
Netgate Services And Support
Interfaces
WAN 1000baseT <full-duplex> 10.0.2.15 fd17:625c:f037:2:a00:27ff:fefc:8147
LAN 1000baseT <full-duplex> 192.168.56.108
Type here to search
Tem...
10:56 AM 4/12/2026
Right Ctrl

Ports enabled:

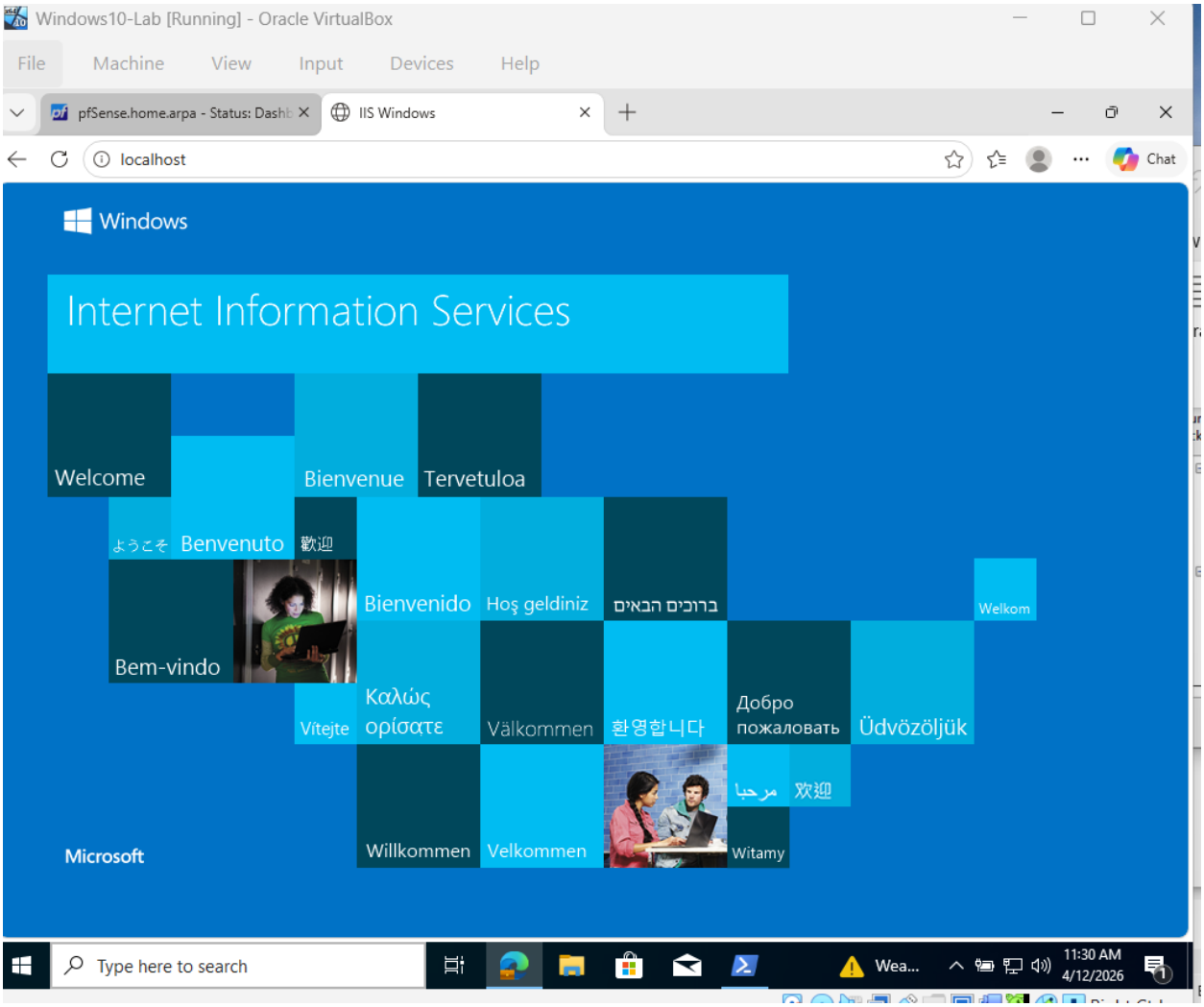
Administrator: Windows PowerShell

```
Name                          : {020284c1-9c45-4208-b2d4-272f9d0675f8}
DisplayName                   : SQL Port 1433
Description                   :
DisplayGroup                  :
Group                         :
Enabled                       : True
Profile                       : Any
Platform                      : {}
Direction                     : Inbound
Action                        : Allow
EdgeTraversalPolicy           : Block
LooseSourceMapping            : False
LocalOnlyMapping              : False
Owner                         :
PrimaryStatus                 : OK
Status                        : The rule was parsed successfully from the store. (65536)
EnforcementStatus             : NotApplicable
PolicyStoreSource             : PersistentStore
PolicyStoreSourceType         : Local
RemoteDynamicKeywordAddresses :
PolicyAppId                   :


PS C:\Windows\system32> New-NetFirewallRule -DisplayName "HTTPS Port 443" -Direction Inbound -Protocol TCP -LocalPort 44
3 -Action Allow


Name                          : {70bb4c48-98c7-4d57-8a9f-a0ce48f0c7b7}
DisplayName                   : HTTPS Port 443
Description                   :
DisplayGroup                  :
Group                         :
Enabled                       : True
Profile                       : Any
Platform                      : {}
Direction                     : Inbound
Action                        : Allow
EdgeTraversalPolicy           : Block
LooseSourceMapping            : False
LocalOnlyMapping              : False
Owner                         :
PrimaryStatus                 : OK
Status                        : The rule was parsed successfully from the store. (65536)
EnforcementStatus             : NotApplicable
PolicyStoreSource             : PersistentStore
PolicyStoreSourceType         : Local
```

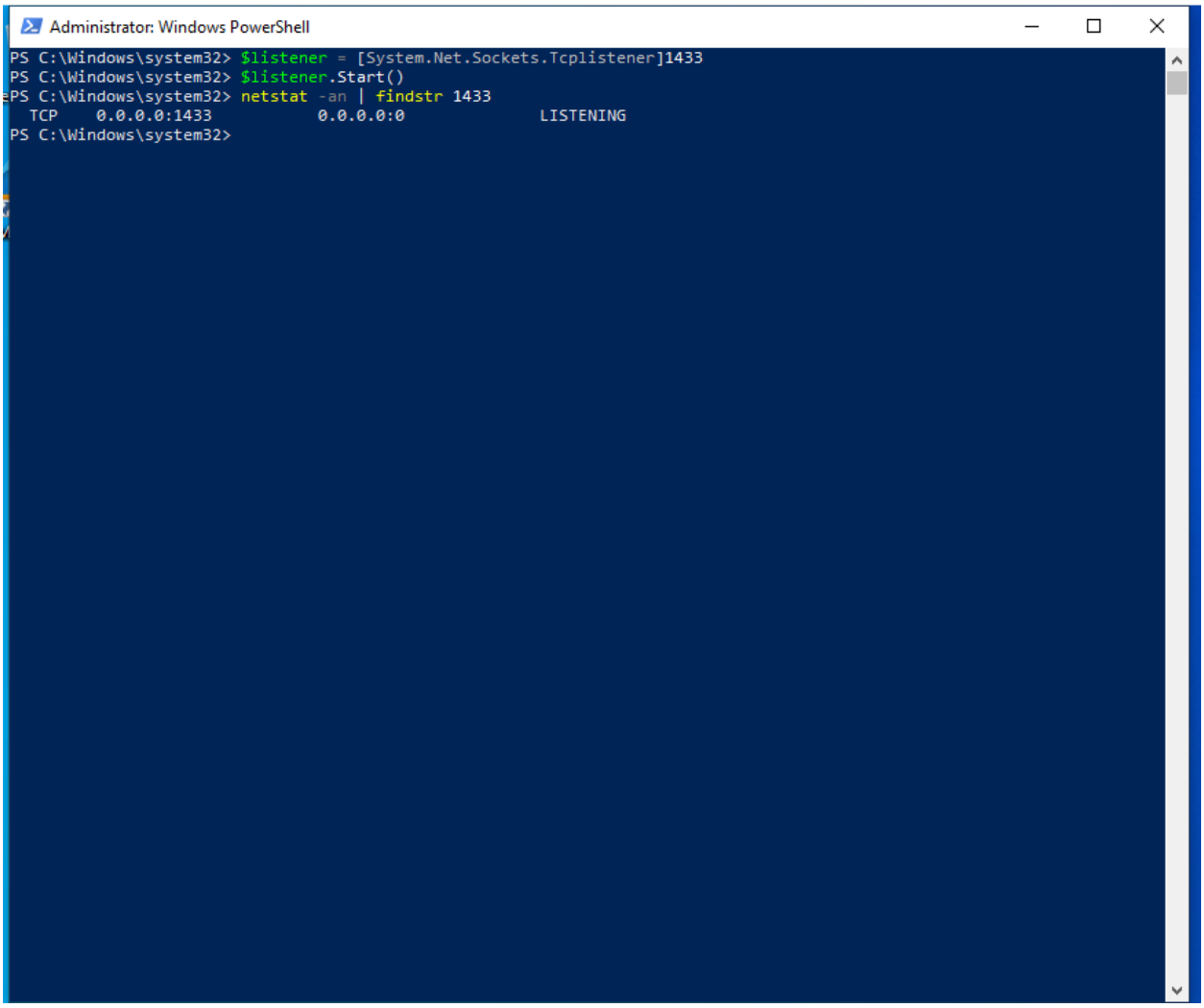
Setting up via control panel:



Opening local host:



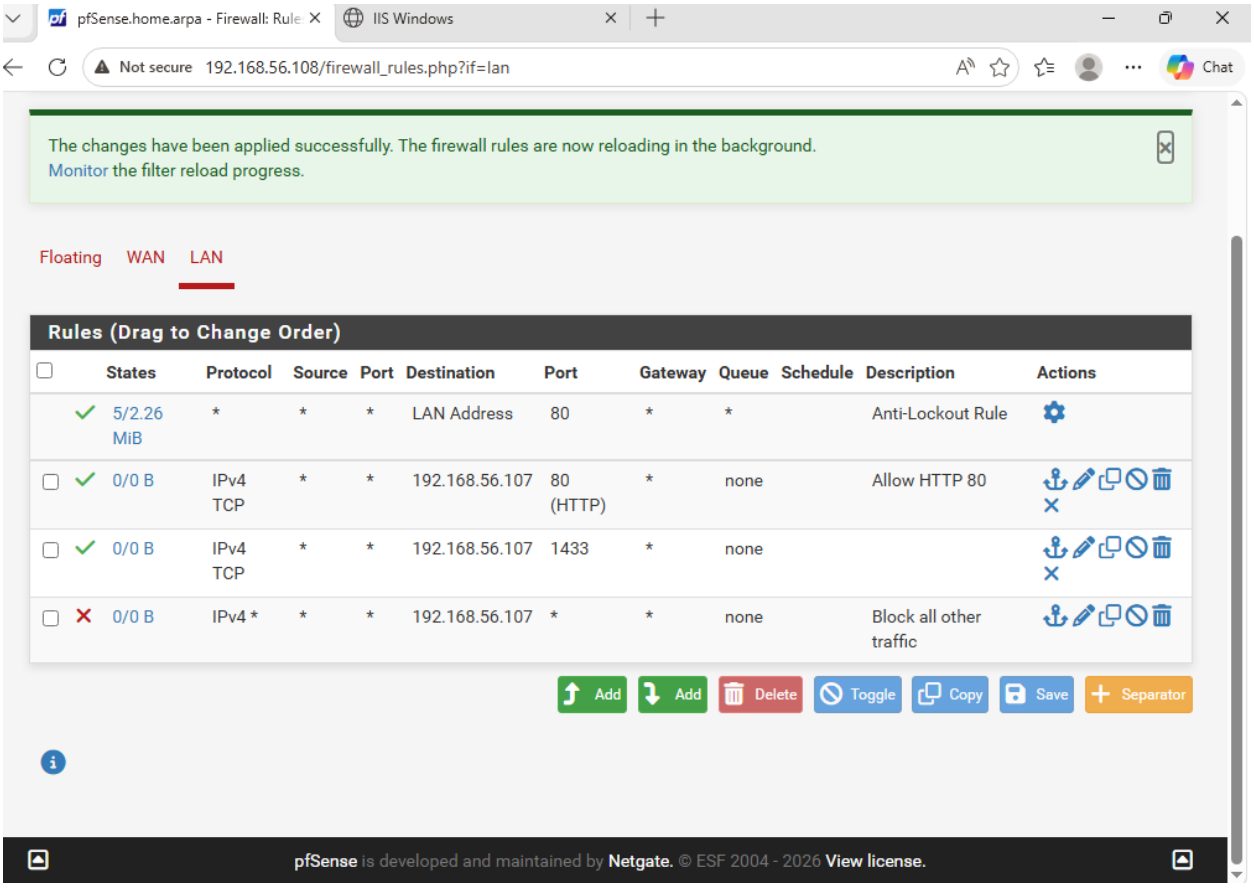
Simulating DB port:



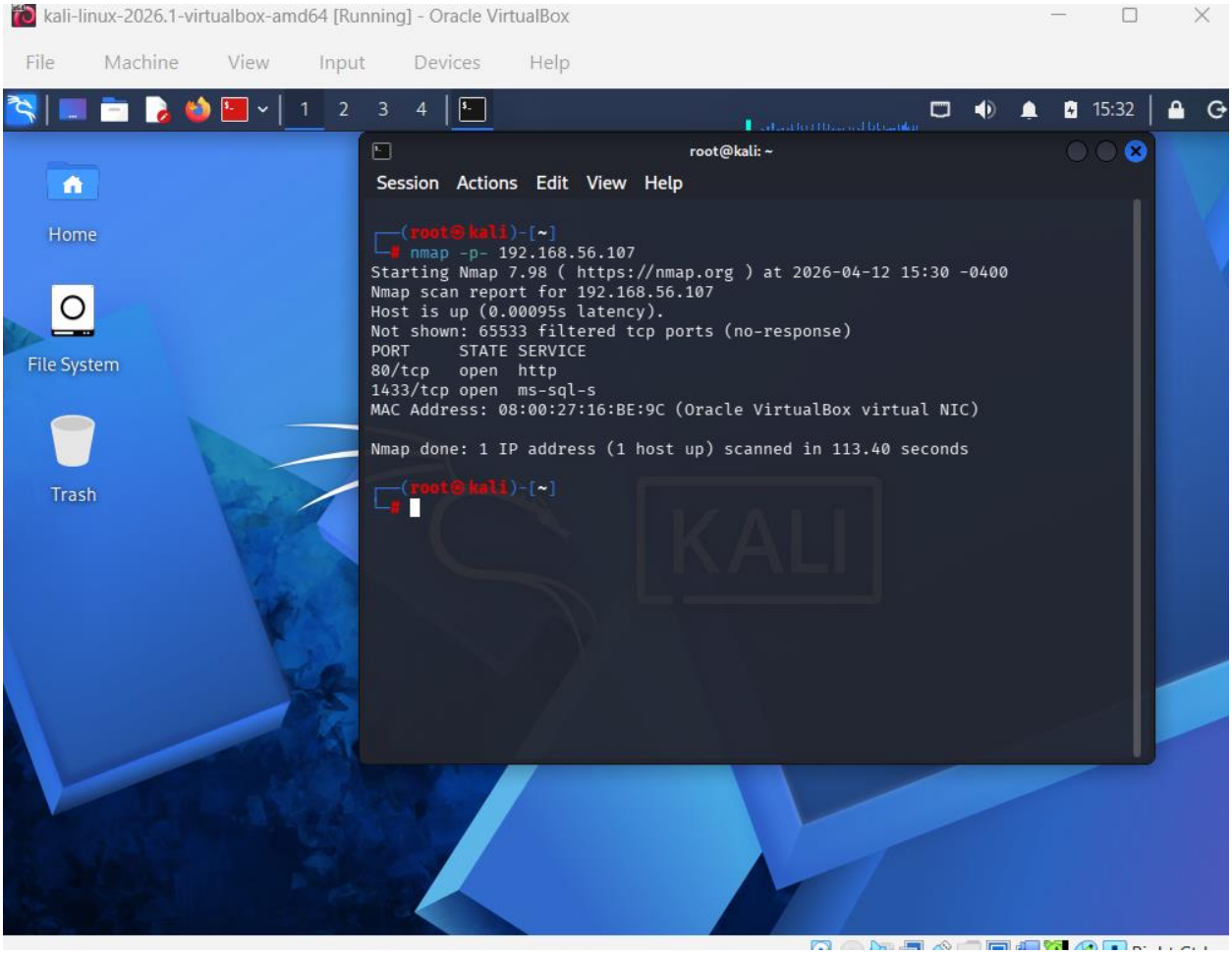
Open Pot 80 in windows firewall:

```
Name                          : {523332c1-fea3-464b-9959-8be6270c1123}
DisplayName                   : HTTP Port 80
Description                   :
DisplayGroup                  :
Group                         :
Enabled                       : True
Profile                       : Any
Platform                      : {}
Direction                     : Inbound
Action                        : Allow
EdgeTraversalPolicy           : Block
LooseSourceMapping            : False
LocalOnlyMapping              : False
Owner                         :
PrimaryStatus                 : OK
Status                        : The rule was parsed successfully from the store. (65536)
EnforcementStatus             : NotApplicable
PolicyStoreSource             : PersistentStore
PolicyStoreSourceType         : Local
RemoteDynamicKeywordAddresses :
PolicyAppId                   :


PS C:\Windows\system32>
```

Firewall rules created:



Nmap scan on Kali:



Conclusion

This lab demonstrated the practical implementation of network security controls using pfSense. A virtual lab environment consisting of Kali Linux, Windows 10, and pfSense was successfully configured to simulate a real-world network. Firewall rules were created to allow specific services, including HTTP (port 80) and SQL (port 1433), while blocking all other unnecessary traffic.

A web server was deployed using IIS on the Windows machine, and a simulated database service was configured to listen on port 1433. Using Nmap from the Kali Linux machine, network scanning confirmed that only the intended ports were accessible, while all other ports were filtered by the firewall.

This exercise reinforced key cybersecurity concepts such as network segmentation, firewall rule configuration, service exposure, and validation through port scanning. The lab highlights the importance of restricting network access to only required services in order to reduce the attack surface and improve overall security posture.

10. Recommendations

- Deploy a real SQL Server instead of simulation for advanced labs
- Implement IDS/IPS tools such as Snort or Suricata on pfSense
- Enable logging and monitoring for better visibility
- Apply similar configurations in enterprise environments